Department of Mathematics

National Institute of Technology Kurukshetra

End Semester Examination

Course Name: Optimization Techniques | Course Code: MAIC 207

Time: 09:30 AM-12:30 PM | Date: 10-12-2025

Use of Calculators is allowed | Total Marks: 50

Do not write anything else except your roll number on the question paper

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1. (4 mark) A company produces two types of belts. Each belt of the first type requires twice as much labor time as the second type. If all belts are of the second type only, the company can produce a total of 500 belts per day. The market limits daily sales of the first and second types to 100 and 200 belts. Assuming that the profits per belt are 8 Rs for type A and 15 Rs for type B, formulate the problem as a linear programming problem.

2. (7 mark) Solve the following LPP using simplex method.

$$\begin{aligned} \text{Maximize } z &= 3x_1 - x_2 \\ \text{subject to } 2x_1 + x_2 &\geq 2 \\ x_1 + 3x_2 &\leq 2 \\ x_1 &\leq 4 \\ x_1, x_2 &\geq 0. \end{aligned}$$

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3. (2 mark) Discuss the major drawbacks or limitations of the simplex method.

4. (7 mark) Solve the following unbalanced transportation problem:

	D_1	D_2	D_3	D_4	D_5	Supply
O_1	4	2	3	2	6	8
O_2	5	4	5	2	1	12
O_3	6	5	4	7	7	14
Demand	4	4	6	8	8	

P.T.O.

5. Answer the following:

(i) (1 mark) Give a reason why transportation problems are solved using specialized algorithms (like VAM, MODI, etc.) instead of the simplex method.

(ii) (2 mark) Why does the MODI (or UV) method require forming a closed loop when improving a solution? Give a brief reasoning.

6. (3 mark) Find the local maxima, local minima, and saddle points (if any) of the function

$$f(x, y) = \frac{1}{x^2 + y^2 - 1}.$$

7. (5 mark) Your firm has been asked to design a storage tank for liquid petroleum gas. The customer's specifications call for a cylindrical tank with hemispherical ends, and the tank is to hold 8000 m^3 of gas. The customer also wants to use the smallest amount of material possible in building the tank. What radius and height do you recommend for the cylindrical portion of the tank?

8. (4 mark) Is the following function convex? Justify your answer.

$$f(x_1, x_2, x_3) = x_1 x_2 + 4{x_1}^2 + {x_2}^2 + 2{x_3}^2 - 10 x_1 x_3.$$

9. Answer the following with proper justifications.

(i) (4 mark) Write a general algorithm structure to solve a general unconstrained optimization problem.

(ii) (3 mark) What is the difference between an exact and an inexact line search?

(iii) (3 mark) What is the descent direction in the steepest descent and Newton's Method.

(iv) (2 mark) What are the drawbacks of the Newton's method and suggest some alternates to overcome these drawbacks.

(v) (3 mark) Convert a general constrained optimization problem with equality and inequality constraints into unconstrained optimization problem and suggest some algorithms to solve the constrained problem.

⋆ ⋆ End of Question Paper ⋆ ⋆

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